

The Emergent Spacetime Rendering Engine: Entropic Gravity Gradients, MERA Isometry Constraints, and Biological Wetware Compilation

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Target Track: *Computational Quantum Gravity, Tensor Networks, & Information-Theoretic Physics*

Abstract

We present a unified systems-theoretic architecture that models emergent 3D bulk spacetime ($g_{\mu\nu}$) as a holographic projection compiled from a discrete 2D boundary update loop ($B_{n+1} = F(B_n)$) running at the Planck frequency ($1/t_P$). By mapping physical phenomena directly onto computational mechanics—rest mass as a projection update tax functional ($\mathcal{C}[B_n]$), wavefunction collapse as lazy evaluation, and quantum entanglement as shared boundary memory pointers—this framework bridges quantum field theory, general relativity, and biophysics.

In this work, we present two primary theoretical and numerical breakthroughs:

1. **Entropic Gravitational Tax Gradients:** We formally derive Newton’s inverse-square gravitational force ($F = -Gm_1m_2/r^2$) as the negative spatial gradient of boundary update tax optimization ($\vec{F} = -\nabla\mathcal{C}_{12}$). Numerical simulations (v20.0) demonstrate that as 3D field distortions overlap, merged Multiscale Entanglement Renormalization Ansatz (MERA) boundary trees reduce total computational update tax from 1,495.00 to 1,451.51 units, generating an attractive entropic force curve.
2. **Biological Wetware Compilation & Evolutionary Profiling:** We establish a formal isomorphism between biological form instantiation and holographic rendering. Linear quaternary genetic code (A, C, T, G) compiled via ribosomal machinery folds into 3D nano-architecture to minimize MERA rendering tax. Multi-generational evolutionary simulations (v19.0) show that natural selection operates as an automated substrate profiler, refactoring genetic source code over 100 generations to achieve an **−89.57% rendering tax reduction** ($\Delta G = -62.6$ kJ/mol).

Finally, we report updated predictions across three empirical test beds: Planck-scale cosmic ray dispersion ($\Delta t = 421.52$ ms at 100 GeV over 500 Mpc), transverse holographic noise ($\delta L = 0.254$ fm across 4.0 km interferometer arms), and sub-millimeter non-Newtonian gravity enhancements (+8.21% at $50\ \mu\text{m}$).

Keywords: Holographic Principle, MERA Tensor Networks, Entropic Gravity, Computational Biophysics, Discrete Spacetime Substrate.